

DATA ANALYSIS

Data analysis is the processes of examining, cleaning, transforming, and modelling raw data to discover actionable insights, trends, and patterns that support decision making.

STEPS OF DATA ANALYSIS

➤ **Data Collection –**

The data collection means gathering data from different source in a systematic way.

➤ **Data Inspection –**

It involves examining the collected data to understands its structure and quality. Analytics check data types, identify missing values, and detect errors or inconsistencies.

➤ **Data Cleaning –**

It very important process in the data analytics. Data cleaning is the process of removing errors and making the data suitable for analysis. It includes handling missing values, removing duplicates and correcting inaccurate data.

➤ **Data Transformation –**

It involves converting data into a suitable format for analysis. This includes normalization scaling and encoding categorical variables.

➤ **Data Exploration –**

Its used to analysis the data and discover patterns, trends and relationships.

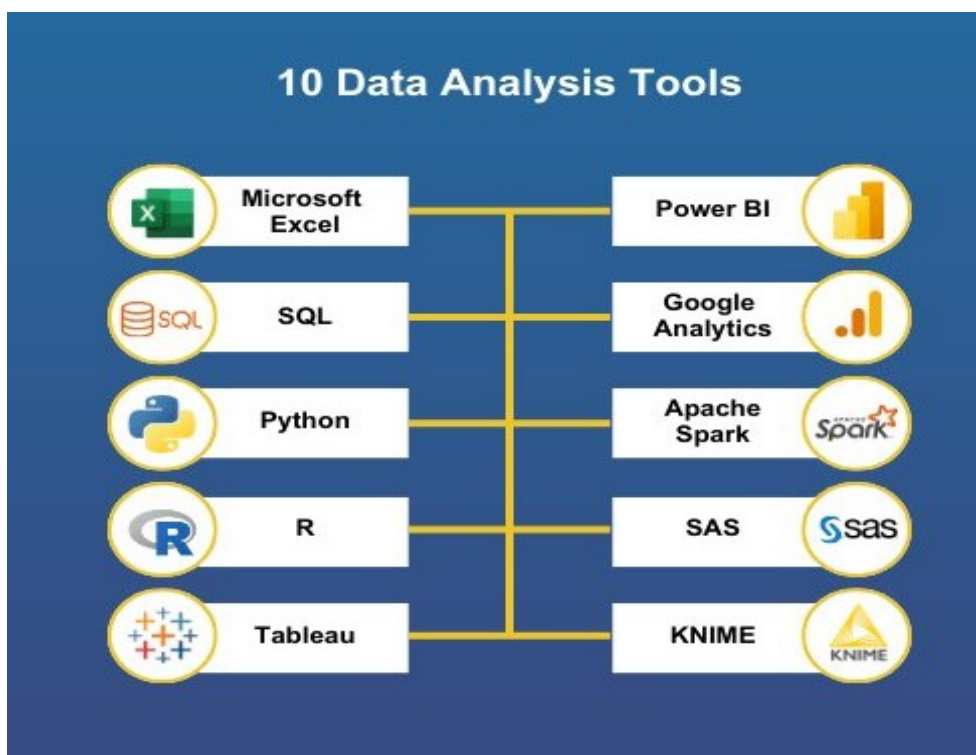
➤ **Data Visualization –**

It present data in graphics formats such as charts and graphs. It helps in simplifying complex data and makes it easier to understands patterns and trends.

➤ **Data Presentation –**

The final stage where insights are communicated to stakeholders. This is done through reports, dashboards or presentation.

TOOLS OF DATA ANALYTICS



1. Microsoft Excel

Best for: Basic data analysis, quick calculations, and visualization.

Excel remains a staple in data analysis due to its simplicity and versatility. With built-in functions, pivot tables, and the Data Analysis ToolPak, users can perform statistical analysis, regression, and hypothesis testing. Excel is ideal for small to medium-sized datasets and is often used for quick visualizations through charts and graphs.

2. SQL (Structured Query Language)

Best for: Database management and querying large datasets.

SQL is essential for working with structured databases. It allows users to extract, filter, aggregate, and manipulate large amounts of data efficiently. Analysts rely on SQL to access relational databases, perform complex joins, and retrieve meaningful insights for decision-making. It is particularly useful for businesses managing extensive customer or transactional data.

3. Python

Best for: Advanced analytics, machine learning, and exploratory data analysis.

Python is a powerful programming language with extensive libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn, which facilitate qualitative data analysis, statistical modeling, and predictive analytics. Python is widely used for automation, machine learning applications, and big data processing, making it a preferred choice for data scientists and analysts.

4. R

Best for: Statistical computing and data visualization.

R is a statistical computing tool favored by researchers and data analysts for its robust analytical capabilities. With packages like ggplot2, dplyr, and tidyr, R excels in hypothesis testing, data modeling, and visualizing trends. It is particularly useful for academic research, social sciences, and financial analytics.

5. Tableau

Best for: Interactive data analysis and visualization.

Tableau is a leading business intelligence tool that allows users to create interactive dashboards and reports. It supports data blending, drag-and-drop analytics, and connectivity to multiple data sources. Tableau is widely used by organizations for storytelling through data, making insights more accessible to decision-makers.

6. Power BI

Best for: Enterprise-level business intelligence and data analytics.

Microsoft Power BI provides a comprehensive suite of tools for data modeling, real-time analytics, and interactive reporting. It integrates seamlessly with other Microsoft applications, allowing businesses to leverage cloud-based analytics for trend forecasting and decision-making.

7. Google Analytics

Best for: Web analytics and tracking online data.

Google Analytics is crucial for digital marketers and website owners to monitor traffic, user behavior, and conversion rates. It provides insights into customer journeys, marketing effectiveness, and audience demographics, helping businesses optimize their online presence.

8. Apache Spark

Best for: Big data processing and real-time analytics.

Apache Spark is an open-source framework for distributed computing, built to handle large-scale data processing efficiently. It supports real-time analytics, machine learning, and AI-driven insights. Spark is widely used by tech companies and enterprises dealing with extensive data sets requiring fast processing speeds.

9. SAS (Statistical Analysis System)

Best for: Enterprise-level statistical analysis and predictive modeling.

SAS is a powerful tool used in industries like healthcare, finance, and government for advanced analytics. It supports forecasting, risk analysis, and data mining, enabling organizations to make data-driven strategic decisions.

10. KNIME

Best for: Workflow automation and visual data analysis.

KNIME is an open-source analytics platform that allows users to automate data processing workflows without coding. It supports statistical analysis, machine learning, and data visualization, making it an excellent choice for organizations looking to streamline repetitive data tasks.

